

# The APWG eCrime eXchange (eCX) start-up guide

You may have recently joined the APWG eCX data clearinghouse. This document is the “new user” guide to explain our processes and how to use the eCX efficiently.

The eCX has two access paths: a GUI at <https://ecrimex.net>, and an API at <https://ecrimex.net/api/v1/<module>>.

The eCX data is divided into different data modules for different types of data. This division allows for each module to hold different data elements. For example, the phish module has a ‘URL’ field as that’s kind of important for phishing collectors. The malicious domain module does not have a URL field because the data records hold DNS names.

## The User Process

The overall user experience follows this path:

1. The Invitation. eCX users are sent an invitation by email to join the platform. Clicking on the invitation link allows the new user to setup an eCX account - entering their name, selecting a password, etc. - then clicking OK.
2. User Approval. The system administrators are notified when a user sets up their initial account and approves it. There may be a delay in the user setting up the account and the system admins performing their task, as they also sleep.
3. Initial Module Access. A new user has read access to the phish data module by default. This access is for both the GUI and the API. We suggest that a new user logs into the eCX GUI (<https://ecrimex.net>) to acquaint themselves. Roll-over “modules” (at the top of the page), and select “phish” to view the current data in the phish module.
4. The User Profile. At the top right of the GUI page is your username. Clicking on it will show two options: “Log Out” and “Profile”. Log out will do just that. Your profile shows
  - a. Information about you.
  - b. Information about your organization’s membership status.
  - c. Your API key. The key can be changed by clicking on the icon at the right of the key display.
5. Changing your Password. The easiest way to change your password is to visit <https://ecrimex.net> and click the “forgot your password?” link.
6. If your co-workers move on from your organization, we would appreciate a note letting us know so we can disable their accounts, reducing our password exposure on unused accounts.

## The Modules

The ECX data modules are:

1. Phish. This data set contains URLs of phishing sites and the brand they are impersonating.
2. ReportPhishing. This contains the slightly-processed emails sent by the public to our [reportphishing@apwg.org](mailto:reportphishing@apwg.org) email address. Normally receives 300,000-500,000 monthly.
3. Malicious\_domain. DNS domains that are a fake store front or serving malicious content.
4. Malicious\_IP. IP Addresses that appear to only attack or probe others on the Internet.
5. Cryptocurrency-addresses. Records of crypto WalletIds seen in crimeware.
6. Malicious\_SMS. This is data retrieved from phish or malicious SMS or TXT messages.

## Confidence Factors

Each data record in the eCX is **REQUIRED** to contain a confidence value. The confidence value informs other eCX uses how confident you are of the maliciousness of this data. There are five normal confidence values:

100    The data was verified by a trained analyst.

90     Our real sexy AI or script found the data, and it rarely makes errors.

50     It looked bad, but this is really a guess. Submissions from the public get a 50.

0       Oops. This is a false positive. This value is extremely important as it lets us quickly notify other users to not block or filter this data value.

\*\*\*eCX users may use the GUI or the API to “remark” data records to add to their confidence, or more likely, mark a record as a false positive.\*\*\*

## Requesting read access to a module.

To gain access to a specific module, log into the GUI, select the module of interest, read the description, and click the “Request Access” button. System admins will change your permissions.

## Requesting submission access to a module

Access to a module in the GUI gives you the ability to submit data via the GUI.

Submission via the API requires the submission privilege for that module. We ask that you test and perfect your API scripts using the eCX sandbox before asking for submission rights to the production eCX API. We have learned our lessons.

## Using the API

The Sandbox. There is a sandbox of the eCX that periodically gets data inserted for testing. It has no direct connection to the production environment. It is located at <https://sandbox.ecrimex.net>, and your API key should work just fine there.

For each API request, your API key should reside in an HTTP Authorization header, like "Authentication: Bearer api-key-value"

### Accessing eCX data via the API:

Using curl for access to phish

➤ `curl -k https://sandbox.ecrimex.net/api/v1/phish -H "Authorization: Bearer 0123my-api-key890" -H "Content-Type": "application/json"`

Using a python script for API access.

def get:

```
headers = {  
    "Authorization": f"Bearer " + CONFIG['api_key'],  
    "Content-Type": "application/json", # Specify content type for the request  
}  
  
resp = requests.get("https://sandbox.ecrimex.net/api/v1/phish", headers=headers, verify=False )  
print("API Response: " + str(resp.status_code)+ " " + str(resp.content))
```

Users may also use their own scripts in any programming language for access.

## Documentation Available in the eCX GUI:

User Guide, available under the 'Docs' link.

API Swagger Documentation, available in the "[Swagger API Docs](#)" tab.

## Documentation Available in the APWG git:

There are some sample python3 scripts to access the eCX via the API at:

<https://github.com/APWG/ecx3-api-routines>

## Assistance and Support.

Help is always available via email at [support@ecrimex.net](mailto:support@ecrimex.net). Our support staff tries to respond promptly. Please include the error message or API call you are having issues with for niceness.

There is an announcement mail list to let users know if things are broken. Sign up at <https://members.apwg.org/mailman3/lists/ecx-announce.members.apwg.org/>

There is a new eCX-users mail list, too, at <https://members.apwg.org/mailman3/lists/ecx-users.members.apwg.org/>

As APWG members, you will be added to the members list which is good for general queries and questions about new attackers, techniques, or odd results.

We have a discord server, but we are nt sure what to do with it.